THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(d)** 90°	An angle in a semicircle is a right angle.
2	**(a)** 84	Area $= l \times b$.
3	**(b)** $\frac{1}{2}$	$P(E) = \frac{\text{favourable}}{\text{total}}$ with 6 equally likely outcomes.
4	**(d)** 11	Put $n = 5$: $2 \times 5 + 1 = 11$.
5	**(b)** 7	Radius $= \frac{d}{2}$.
6	**(c)** 46	Perimeter $= 2(l + b)$.
7	**(a)** $\frac{1}{2}$	Two equally likely outcomes.
8	**(b)** 3	Subtract any term from the next term.
9	**(b)** 5	The perpendicular from the centre bisects the chord.
10	**(c)** 36	Perimeter $= 4\times$ side.
11	**(d)** $\frac{1}{2}$	$P = \frac{\text{green balls}}{\text{all balls}}$.
12	**(d)** 22	Add the common difference 5 to the last term.
13	**(d)** 25°	The angle at the centre is double the angle at the circumference.
14	**(c)** 48	Area $= \frac{1}{2}\times$ base $\times$ height.
15	**(a)** $\frac{1}{2}$	Count the multiples of the number.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	Angle at the centre is double that at the circumference (1); Required angle $= \frac{36^\circ}{2} = 18^\circ$ (1).	1+1
17	$C = 2\pi r = 2 \times \frac{22}{7} \times 21$ (1); $C = 132$ cm (1).	1+1
18	Favourable outcomes $= 3$, total outcomes $= 6$ (1); $P = \frac{1}{2}$ (1).	1+1
19	$t_1 = 4$, $t_2 = 7$ (1); $t_3 = 10$, $t_4 = 13$ (1).	1+1
20	Opposite angles are supplementary (1); $\angle D = 180^\circ - 72^\circ = 108^\circ$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Half-chord $= \sqrt{5^2 - 3^2}$ (1); $= \sqrt{16} = 4$ cm (1); Chord $= 2 \times 4 = 8$ cm (1).	1+1+1
22	Area $=$ base $\times$ height (1); $108 = 12 \times h$ (1); $h = 9$ cm (1).	1+1+1
23	Total $= 10$ (1); $P(\text{red}) = \frac{3}{10}$ (1); $P(\text{not red}) = 1 - \frac{3}{10} = \frac{7}{10}$ (1).	1+1+1
24	$t_n = a + (n-1)d$ (1); $t_{10} = 3 + 9 \times 4$ (1); $= 39$ (1).	1+1+1
	OR: Angle at the centre is double that at the circumference (1); $\angle ACB = 24^\circ$ (1); Angles in the same segment are equal (1).	1+1+1
25	$s = \frac{3+4+5}{2} = 6$ (1); $A = \sqrt{s(s-a)(s-b)(s-c)}$ (1); $A = 6$ cm^2 (1).	1+1+1
	OR: $P(4) = \frac{22}{120} = \frac{11}{60}$ (1); $P(\text{not } 4) = 1 - \frac{11}{60}$ (1); $= \frac{49}{60}$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	The perpendicular from the centre bisects the chord (1); Half-chord $= 12$ cm (1); $r = \sqrt{12^2 + 9^2} = \sqrt{225}$ (1); $r = 15$ cm (1); Radius found using the right triangle (1).	1+1+1+1+1
	OR: Perimeter $= 2(40 + 25) = 130$ m (1); Area $= 40 \times 25 = 1000$ m^2 (1); Cost $= 130 \times 12$ (1); $=$ ₹1560 (1); Units stated (1).	1+1+1+1+1
27	Sample space: HH, HT, TH, TT (4 equally likely outcomes) (1); $P(\text{two heads}) = \frac{1}{4}$ (1); $P(\text{at least one head}) = \frac{3}{4}$ (1); $P(\text{no head}) = \frac{1}{4}$ (1); Check: $P(\text{no head}) = 1 - P(\text{at least one head})$ (1).	1+1+1+1+1

	OR: Terms $4, 7, 10, 13, 16$ (1); Constant difference, so an AP (1); $t_{10} = 4 + 9 \times 3$ (1); $= 31$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** 16	Multiply the last term by the common ratio 2.
2	**(d)** 40°	Central angle $=2\times$ angle at the circumference.
3	**(a)** 120	Area $=$ base $\times$ height.
4	**(a)** 0	Impossible events have probability 0; sure events have probability 1.
5	**(a)** 2	Divide a term by the previous term.
6	**(b)** 115°	Opposite angles of a cyclic quadrilateral are supplementary.
7	**(d)** 44	$C=2\pi r$.
8	**(b)** 6	Count all possible outcomes.
9	**(b)** 21	$\frac{n(n+1)}{2}=\frac{6\times7}{2}$.
10	**(d)** diameter	Recall the circle theorems.
11	**(b)** 154	$A=\pi r^2$.
12	**(c)** $\frac{5}{7}$	$P(\text{not } E)=1-P(E)$.
13	**(a)** an AP	Constant difference 2.
14	**(c)** 90°	An angle in a semicircle is a right angle.
15	**(c)** 7	$r=\frac{C}{2\pi}$.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$A=\pi r^2=\frac{22}{7}\times7\times7$ (1); $A=154$ cm^2 (1).	1+1
17	Total balls $=8$ (1); $P=\frac{5}{8}$ (1).	1+1
18	$d=4$ (1); Next terms 22 and 26 (1).	1+1
19	$\angle ACB=90^\circ$ (angle in a semicircle) (1); $\angle ABC=180^\circ-90^\circ-28^\circ=62^\circ$ (1).	1+1
20	$A=\frac{1}{2}\times18\times10$ (1); $A=90$ cm^2 (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	$S_n=\frac{n(n+1)}{2}$ (1); $S=20\times21\div2$ (1); $=210$ (1).	1+1+1
22	Opposite angles are supplementary (1); $\angle C=180^\circ-70^\circ=110^\circ$ (1); $\angle D=180^\circ-85^\circ=95^\circ$ (1).	1+1+1
23	$A=\frac{1}{2}(a+b)h$ (1); $A=\frac{1}{2}\times(10+14)\times6$ (1); $A=72$ cm^2 (1).	1+1+1
24	Multiples of 5: 5, 10, 15, 20, ... (6 numbers) (1); $P=\frac{6}{30}=\frac{1}{5}$ (2).	1+2
	OR: $t_n=ar^{n-1}$ (1); $t_6=1\times2^5$ (1); $=32$ (1).	1+1+1
25	Half-chord $=\sqrt{5^2-4^2}$ (1); $=\sqrt{9}=3$ cm (1); Chord $=2\times3=6$ cm (1).	1+1+1
	OR: Area $=$ base $\times$ height (1); $120=15\times h$ (1); $h=8$ cm (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	Angle in a semicircle is 90° so $\angle ACB=\angle ADB=90^\circ$ (1); $\angle ABC=90^\circ-25^\circ=65^\circ$ (1); $\angle ABD=90^\circ-30^\circ=60^\circ$ (1); $\angle CBD=125^\circ$ is not needed here (1); Sum check: $\angle ABC+\angle ABD=125^\circ$ (1).	1+1+1+1+1
	OR: $C=2\pi r=88$ m (1); $A=\pi r^2=616$ m^2 (1); Cost $=88\times15$ (1); $=$ ₹1320 (1); Units stated (1).	1+1+1+1+1
27	Total outcomes $=6$ (1); $P(\text{an even number})=\frac{3}{6}=\frac{1}{2}$ (1); $P(\text{a number greater than } 4)=\frac{2}{6}=\frac{1}{3}$ (1); Favourable for 'or': 4 outcomes (1); $P=\frac{4}{6}=\frac{2}{3}$ (1).	1+1+1+1+1
	OR: $S_n=\frac{n}{2}[2a+(n-1)d]$ (1); $S=10/2\times[4+27]$ (1); $=5\times31$ (1); $S_{10}=155$ (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | Class 9 Maths | Maximum Marks: 50

SECTION — A (Objective Type) — Answers — 15 × 1 = 15

Q.	Correct Option	Brief Reasoning
1	**(b)** $\frac{23}{50}$	Relative frequency $= \frac{\text{times it occurred}}{\text{number of trials}}$.
2	**(b)** 36	The sequence is n^2; $6^2 = 36$.
3	**(a)** 9	Radius $= \frac{d}{2}$.
4	**(b)** 135	Area $= l \times b$.
5	**(a)** $\frac{1}{3}$	$P(E) = \frac{\text{favourable}}{\text{total}}$ with 6 equally likely outcomes.
6	**(a)** 14	Put $n = 4$: $3 \times 4 + 2 = 14$.
7	**(b)** 8	The perpendicular from the centre bisects the chord.
8	**(c)** 62	Perimeter $= 2(l + b)$.
9	**(b)** $\frac{1}{2}$	Two equally likely outcomes.
10	**(c)** 5	Subtract any term from the next term.
11	**(c)** 32°	The angle at the centre is double the angle at the circumference.
12	**(c)** 52	Perimeter $= 4\times$ side.
13	**(b)** $\frac{1}{3}$	$P = \frac{\text{green balls}}{\text{all balls}}$.
14	**(a)** 25	Add the common difference 4 to the last term.
15	**(c)** 70°	Central angle $= 2\times$ angle at the circumference.

SECTION — B (Very Short Answer) — Answers — 5 × 2 = 10

Q.	Expected Answer / Value Points	Marks
16	$P(\text{head}) = \frac{33}{60} = \frac{11}{20}$ (1); $P(\text{tail}) = \frac{27}{60} = \frac{9}{20}$ (1).	1+1
17	$r = 5$ (1); Fourth term $= 2 \times 5^3 = 250$ (1).	1+1
18	Angle at the centre is double that at the circumference (1); Required angle $= \frac{84^\circ}{2} = 42^\circ$ (1).	1+1
19	$C = 2\pi r = 2 \times \frac{22}{7} \times 42$ (1); $C = 264$ cm (1).	1+1
20	Favourable outcomes $= 4$, total outcomes $= 6$ (1); $P = \frac{2}{3}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 5 × 3 = 15

Q.	Expected Answer / Value Points	Marks
21	Total $= 9$ (1); $P(\text{red}) = \frac{5}{9}$ (1); $P(\text{not red}) = 1 - \frac{5}{9} = \frac{4}{9}$ (1).	1+1+1
22	$t_n = a + (n-1)d$ (1); $t_{10} = 5 + 9 \times 2$ (1); $= 23$ (1).	1+1+1
23	Angle at the centre is double that at the circumference (1); $\angle ACB = 33^\circ$ (1); Angles in the same segment are equal (1).	1+1+1
24	$s = \frac{5+12+13}{2} = 15$ (1); $A = \sqrt{s(s-a)(s-b)(s-c)}$ (1); $A = 30$ cm^2 (1).	1+1+1
	OR: $P(6) = \frac{8}{60} = \frac{2}{15}$ (1); $P(\text{not } 6) = 1 - \frac{2}{15}$ (1); $= \frac{13}{15}$ (1).	1+1+1
25	$S_n = \frac{n(n+1)}{2}$ (1); $S = 30 \times 31 \div 2$ (1); $= 465$ (1).	1+1+1
	OR: Opposite angles are supplementary (1); $\angle C = 180^\circ - 100^\circ = 80^\circ$ (1); $\angle D = 180^\circ - 75^\circ = 105^\circ$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
26	The perpendicular from the centre bisects the chord (1); Half-chord $= 9$ cm (1); $r = \sqrt{9^2 + 12^2} = \sqrt{225}$ (1); $r = 15$ cm (1); Radius found using the right triangle (1).	1+1+1+1+1
	OR: Perimeter $= 2(60 + 35) = 190$ m (1); Area $= 60 \times 35 = 2100$ m^2 (1); Cost $= 190 \times 15$ (1); $=$ ₹2850 (1); Units stated (1).	1+1+1+1+1
27	Total balls $= 15$ (1); $P(\text{red}) = \frac{4}{15}$, $P(\text{blue}) = \frac{1}{3}$ (1); $P(\text{green}) = \frac{2}{5}$ (1); Sum $= \frac{4+5+6}{15} = 1$ (1); Hence verified (1).	1+1+1+1+1
	OR: Terms $7, 12, 17, 22, 27$ (1); Constant difference, so an AP (1); $t_{10} = 7 + 9 \times 5$ (1); $= 52$ (2).	1+1+1+2

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